Session 3. Quantitative variables.

Pre course: Statistics.

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Agenda

- Quantitative variables.
- Descriptive statistics.
- Steam and leaf displays.
- Distribution viz – dot plots.
- Distribution viz – histograms.
- Exercises.

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Types of variables.

- Categorical variables (qualitative).
  - Ordinal variables (order does matter).
  - Nominal variables (order **does not** matter).
    - Binary (or dummy or 0,1) included.

Qualitative variables.

- Discrete variables.
- Continuous variables.

Quantitative variables.

Quantitative variables.

- Discrete and continuous variables.
- We can do all calculations with them.
- In the case of qualitative variables, the options are more reduced.

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Descriptive statistics.

- Central tendency measures:
  - mean,
  - median,
  - mode.
- Dispersion:
  - range,
  - standard deviation,
  - variance.

Descriptive statistics.

- Central tendency measures:
  - Mean.
  - Median.
- Dispersion:
  - Range.

3	12	17	20	22	22	28	33	34	35	35	37	42	46	58

Descriptive statistics.

- Central tendency measures:
  - Mean: 29,6.
  - Median: 33.
- Dispersion:
  - Range: 3 - 58.

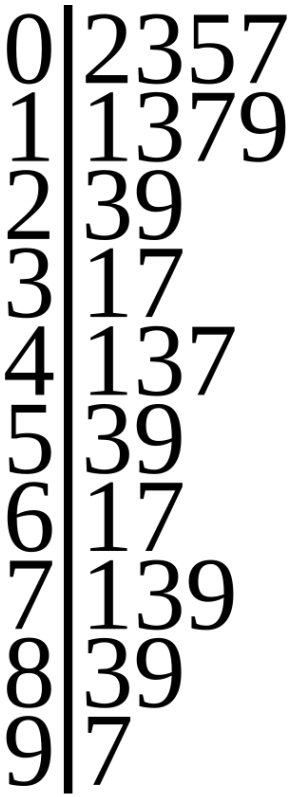
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Steam and leaf displays.

- Organize and visualize qualitative data distributions.
- Separate a steam (like the tens) and the leaf (like the units).
- Keep numerical information (in contrast with bar plots or other visualizations).
- Allow us to see patterns better than in a table.
- First step towards distribution visualizations.

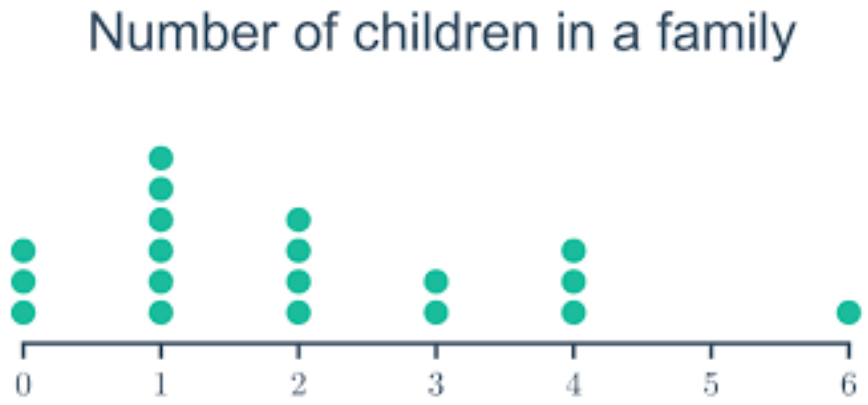


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Dot plots.

- More visual than the steam and leaf displays.
- They also allow us to understand the distribution of the data within its range.
- Ideal for small data sets.
- For bigger data sets other tools fit better.



Before our dot plot.

- We will organize our quantitative variable into ranges

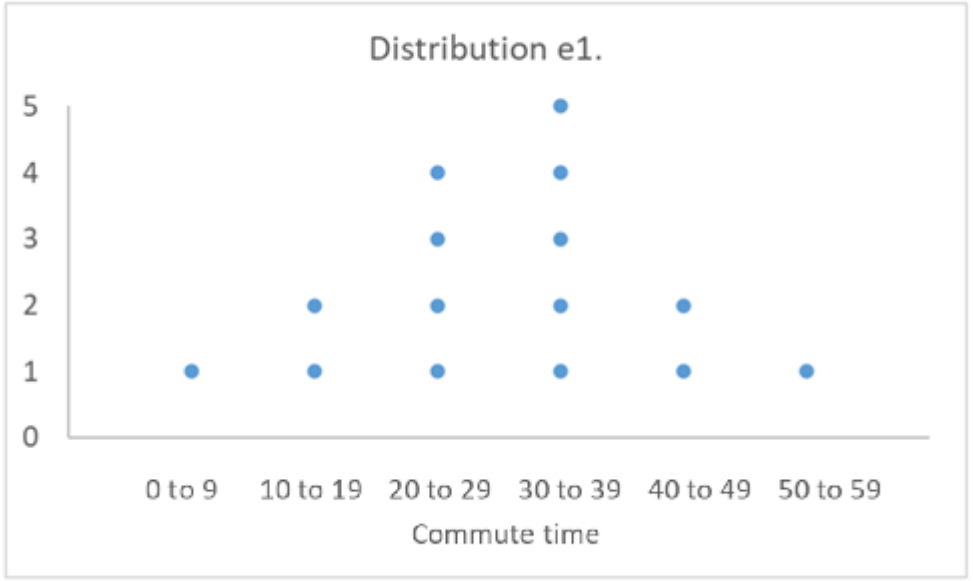
3	12	17	20	22	22	28	33	34	35	35	37	42	46	58
4	3	5	6	14	18	22	28	33	35	48	51	53	81	85

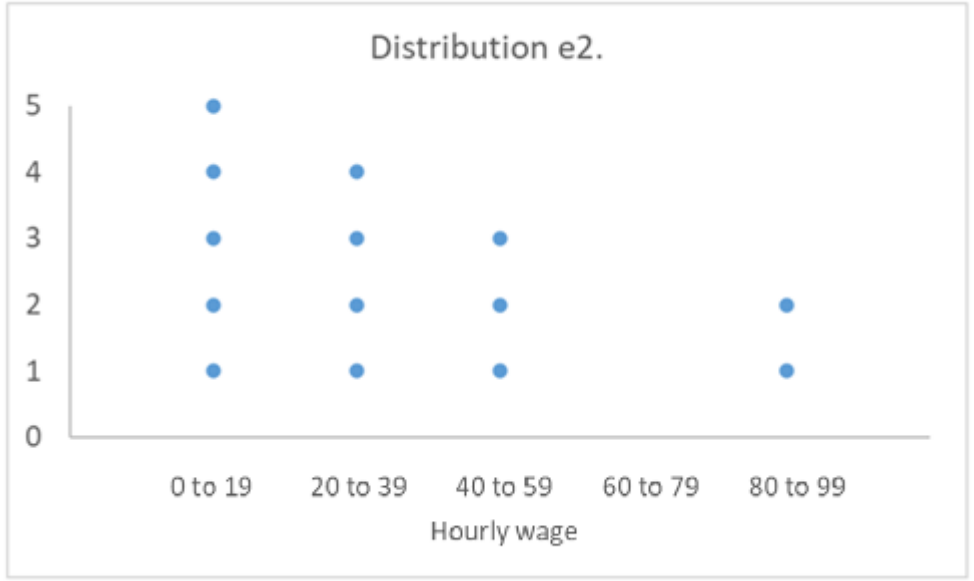
Before our dot plot.

- We will organize our quantitative variable into ranges.
- The first dot plot has six 10 units-wide ranges.
- The second has five 20 units-wide ranges.

3	12	17	20	22	22	28	33	34	35	35	37	42	46	58
4	3	5	6	14	18	22	28	33	35	48	51	53	81	85

Dot plots.

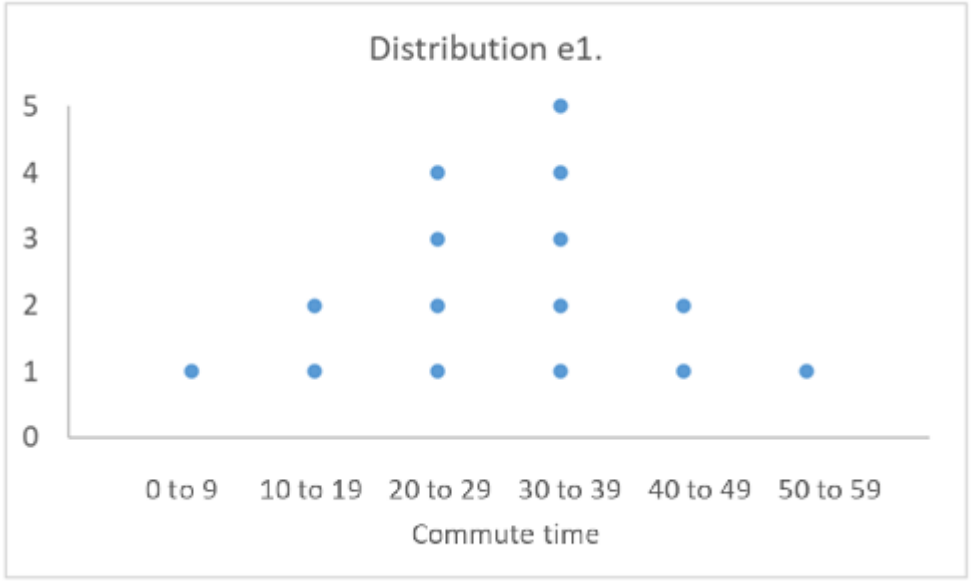

Distribution e1.
5
4
3
2
1
0
0 to 9
10 to 19
20 to 29
30 to 39
40 to 49
50 to 59
Commute time

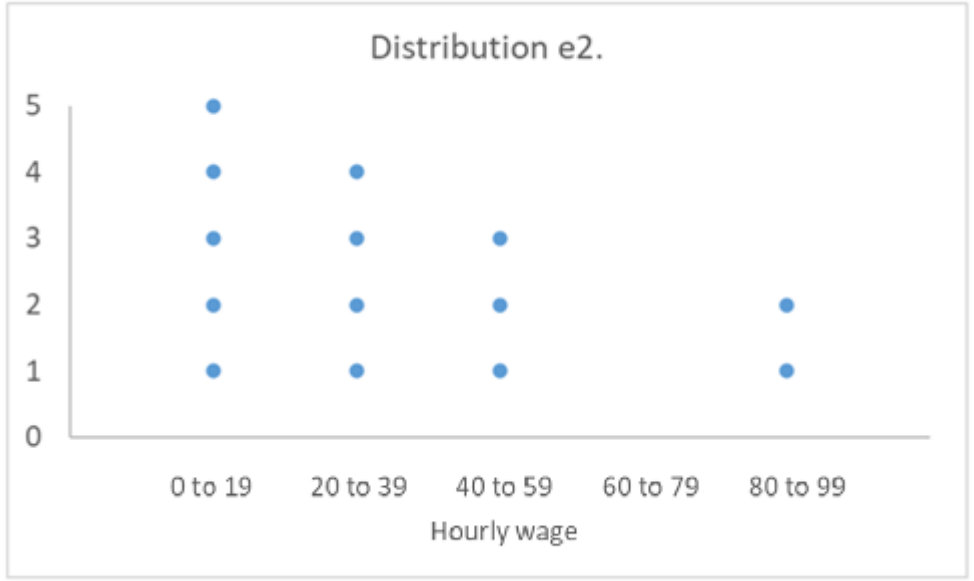

Distribution e2.
5
4
3
2
1
0
0 to 19
20 to 39
40 to 59
60 to 79
80 to 99
Hourly wage

Agenda

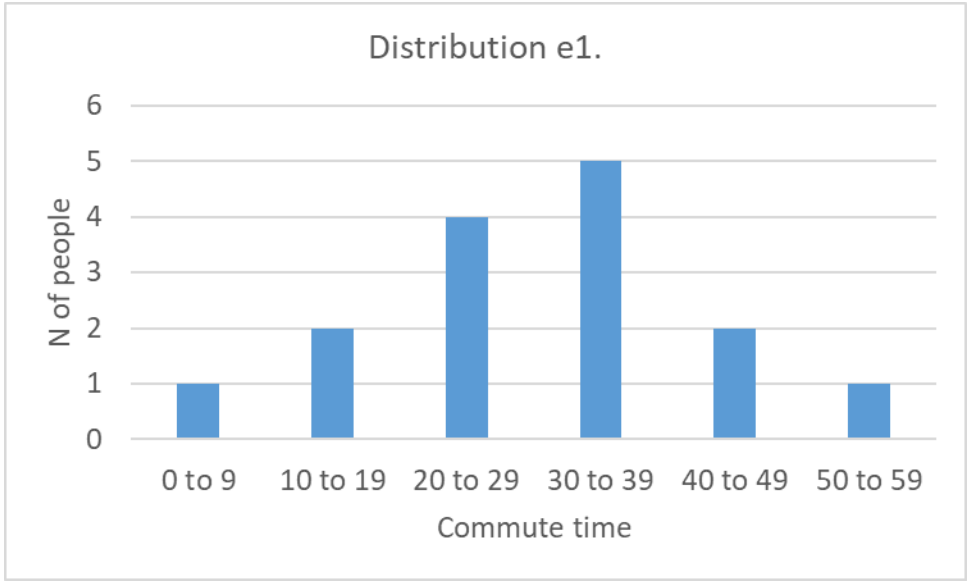
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- **Distribution viz – histograms.**
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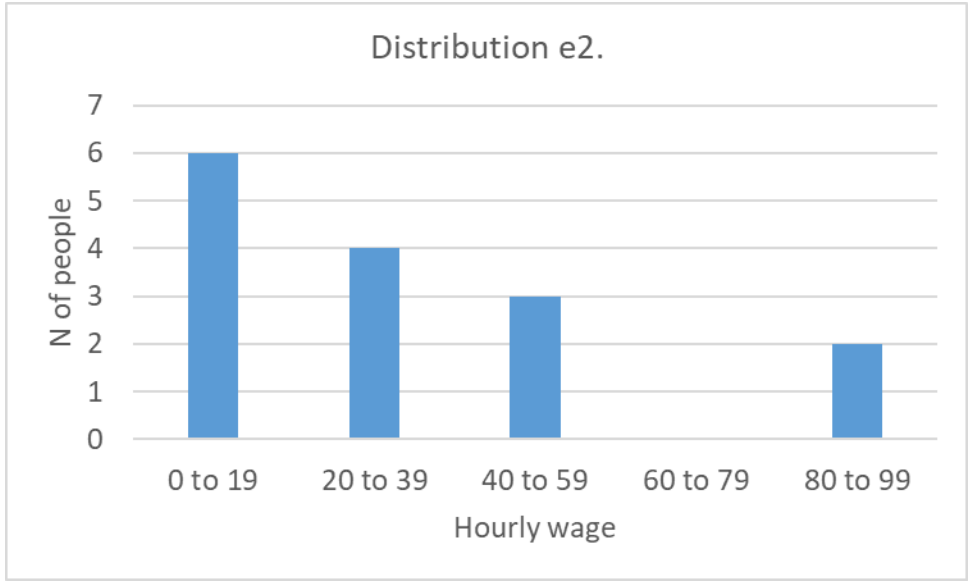
Dot plots.


Distribution e1.
5
4
3
2
1
0
0 to 9
10 to 19
20 to 29
30 to 39
40 to 49
50 to 59
Commute time


Distribution e2.
5
4
3
2
1
0
0 to 19
20 to 39
40 to 59
60 to 79
80 to 99
Hourly wage

Histograms.


Distribution e1.
N of people
6
5
4
3
2
1
0
0 to 9
10 to 19
20 to 29
30 to 39
40 to 49
50 to 59
Commute time


Distribution e2.
N of people
7
6
5
4
3
2
1
0
0 to 19
20 to 39
40 to 59
60 to 79
80 to 99
Hourly wage

Histograms.

- X axis: continuous variable in ranges or bins (equal width intervals) used to group data.
- Y axis: can be number of people but also proportions.
- Excellent for spotting the shape of the distribution.
- Best for large datasets.
- They group values into ranges rather than showing every individual data point.

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Exercises.

- Review the skewness.
- How does the mean work on each graph?
- What else can we say when comparing them?